

Second-Order ODE with a Regular Singular Point at $t = 0$

1 Problem and hypotheses.

Consider

$$t^2 u''(t) + t a(t) u'(t) + b(t) u(t) = 0, \quad (1)$$

where a, b are real-analytic on $|t| < R_0$:

$$a(t) = \sum_{k=0}^{\infty} a_k t^k, \quad b(t) = \sum_{k=0}^{\infty} b_k t^k.$$

Dividing by t^2 yields $u'' + \frac{a(t)}{t} u' + \frac{b(t)}{t^2} u = 0$, so $t = 0$ is a *regular singular point* (Fuchsian), because $t \frac{a(t)}{t} = a(t)$ and $t^2 \frac{b(t)}{t^2} = b(t)$ are analytic at 0. On any closed subinterval of $(0, R_0)$ or $(-R_0, 0)$, the coefficients are analytic, hence the usual existence–uniqueness theory applies.

2 Indicial equation and Frobenius ansatz.

Seek a solution in a punctured neighborhood of 0 of the form

$$u(t) = t^r \sum_{n=0}^{\infty} c_n t^n, \quad c_0 \neq 0.$$

Then

$$u'(t) = t^{r-1} \sum_{n=0}^{\infty} (n+r) c_n t^n, \quad u''(t) = t^{r-2} \sum_{n=0}^{\infty} (n+r)(n+r-1) c_n t^n.$$

Substitute into (1) and collect the coefficient of t^{r+n} :

$$0 = (n+r)(n+r-1) c_n + \sum_{j=0}^n a_j (n-j+r) c_{n-j} + \sum_{j=0}^n b_j c_{n-j}.$$

Define the *indicial polynomial*

$$P(x) = x(x-1) + a_0 x + b_0 = (x-r_1)(x-r_2),$$

with roots $r_1, r_2 \in \mathbb{C}$ (for real a_0, b_0 , either both real or a complex conjugate pair). Note the *Fuchs relations*

$$r_1 + r_2 = 1 - a_0, \quad r_1 r_2 = b_0.$$

At $n = 0$, the equation becomes $P(r) c_0 = 0$, hence the *indicial equation* $P(r) = 0$.

For $n \geq 1$, isolate the $j = 0$ terms and obtain the *recursion* (valid whenever $P(r+n) \neq 0$):

$$c_n = -\frac{1}{P(r+n)} \sum_{j=1}^n [a_j (r+n-j) + b_j] c_{n-j}, \quad n \geq 1. \quad (2)$$

Given $r \in \{r_1, r_2\}$ and $c_0 \neq 0$, this determines uniquely all c_n for which the denominators $P(r+n)$ do not vanish.

3 Convergence of Frobenius series.

Let $R = \min\{R_a, R_b\}$ be a common radius of convergence of a, b at 0. There exists $0 < R^* \leq R$ such that for any indicial root r with the nonresonance condition $P(r + n) \neq 0$ for all $n \geq 1$, the series $\sum_{n \geq 0} c_n t^n$ defined by (2) converges absolutely and uniformly on $|t| \leq \rho < R^*$.

Sketch of proof (majorants). Choose $\rho < R$ and bounds $\sum_{k \geq 0} |a_k| \rho^k \leq A$, $\sum_{k \geq 0} |b_k| \rho^k \leq B$. Also choose $\delta > 0$ and N so that $|P(r + n)| \geq \delta(n + 1)^2$ for $n \geq N$ (possible because P is quadratic and $P(r + n) \sim n^2$ as $n \rightarrow \infty$). Then by induction one shows $|c_n| \leq M \rho^{-n}$ for some M depending on A, B, δ and the finite set $\{c_0, \dots, c_{N-1}\}$, using (2) and the Cauchy product inequality. Hence the power series has positive radius of convergence. $\square$

Remark 3.1. Observe that the radius of convergence is true only for the series $\sum_{k \geq 0} c_k t^k$. This is unrelated to whether this series (multiplied by some power of t solves an ODE or not).

4 Classification of local solutions near $t = 0$ (on $t > 0$).

Let $\operatorname{Re} r_1 \geq \operatorname{Re} r_2$ be the indicial roots. We have the following cases:

4.1 Case (A): Nonresonant distinct real roots ($r_1 - r_2 \notin \mathbb{Z}$).

There exist two linearly independent solutions

$$u_j(t) = t^{r_j} \sum_{n=0}^{\infty} c_n^{(j)} t^n, \quad c_0^{(j)} \neq 0, \quad j = 1, 2,$$

convergent for $0 < t < R^*$. These form a fundamental system on $(0, R^*)$.

4.2 Case (B): Double root ($r_1 = r_2 =: r_*$).

There is a Frobenius solution

$$u_1(t) = t^r \sum_{n=0}^{\infty} c_n t^n,$$

and a second solution of the form

$$u_2(t) = u_1(t) \log t + t^r \sum_{n=0}^{\infty} d_n t^n.$$

Again the coefficients d_n are obtained by a recursion similar to (2), with a modified equation at the resonant index. Below is a clean motivation and derivation of the $\log t$ term that uses *differentiation with respect to the exponent r* . The key observation is that the *operator* in (1) does not depend on r ; only the Frobenius ansatz does.

Step 1: Construct the usual solution for r_* . The equation (2) defines

$$U(t; r_*) = t^{r_*} \Phi(t; r_*), \quad \Phi(t; r_*) = \sum_{n=0}^{\infty} c_n(r_*) t^n, \quad c_0(r_*) = 1,$$

as a *bona fide* local solution of (1) on $0 < |t| < R$, where the coefficients $c_n(r_*)$ are obtained by (2). The recursion is algebraic-rational in r and since there are no resonances $P(r + n) = 0$, depends analytically on r by inspection.

Crucially, the *differential operator*

$$\mathcal{L}[u] := t^2 u'' + t a(t) u' + b(t) u$$

has no r -dependence.

Step 2: Differentiate with respect to r . Fix t with $0 < |t| < R$ and differentiate the identity $\mathcal{L}[U(\cdot; r)] = 0$ with respect to r and evaluate it at r_* :

$$\mathcal{L}[\partial_r U(\cdot; r_*)] = 0,$$

because derivatives commute¹.

Now compute $\partial_r U$ explicitly using the Frobenius form:

$$\partial_r U(t; r)|_{r=r_*} = \partial_r(t^r \Phi(t; r))|_{r=r_*} = t^r \log t \Phi(t; r)|_{r=r_*} + t^r \partial_r \Phi(t; r)|_{r=r_*}.$$

Thus,

$$\partial_r U(t; r)|_{r=r_*} = U(t; r_*) \log t + t^{r_*} \Psi(t; r_*), \quad \Psi(t; r_*) := \partial_r \Phi(t; r_*).$$

Linear independence. For $r \neq r_*$ the two functions $U(\cdot; r)$ and $U(\cdot; \bar{r})$ (or $U(\cdot; r)$ and $U(\cdot; r - \epsilon)$) form a fundamental system on $0 < |t| < R$. Dividing the difference quotient

$$\frac{U(\cdot; r) - U(\cdot; r_*)}{r - r_*}$$

and letting $r \rightarrow r_*$ shows that $\tilde{U}_2$ is not a scalar multiple of U_1 (its Wronskian with U_1 is the nonzero limit of the Wronskian quotients). Hence $\{U_1, \tilde{U}_2\}$ is a fundamental system near $t = 0$.

4.2.1 Differentiating with respect to r term by term; justification

The coefficients $c_n(r)$ satisfy the recursion, for $n \geq 1$,

$$c_n(r) = -\frac{1}{P(r+n)} \sum_{j=1}^n (a_j(r+n-j) + b_j) c_{n-j}(r).$$

For $n \geq 1$, the denominators $P(r+n)$ are nonzero when r is close to r_* (since the only root of P is at $r = r_*$ itself).

By induction on n :

- Each $c_n(r)$ is a rational function of r , hence analytic near r_* .
- For any fixed $\rho < R$, bounds of the form

$$|c_n(r)| \leq M \rho^{-n}, \quad r \in U,$$

hold uniformly for r in a neighborhood U of r_* . This follows because $P(r+n) \sim n^2$ as $n \rightarrow \infty$, and the recursion ensures a geometric bound (see the argument in Section 3).

¹This requires a slight justification, which you are welcome to look up!

Hence $\Phi(t; r) = \sum_{n \geq 0} c_n(r) t^n$ and $\Psi(t; r) = \sum_{n \geq 0} \partial_r c_n(r) t^n$ converge uniformly on $|t| \leq \rho < R$, uniformly for $r \in U$, so we may differentiate term-by-term²:

$$\partial_r \Phi(t; r) = \sum_{n=0}^{\infty} \partial_r c_n(r) t^n.$$

This series converges uniformly on $|t| \leq \rho < R$, with the same radius of convergence as Φ , because the recursion for $\partial_r c_n(r)$ has the same denominators $P(r+n) \sim n^2$ and thus satisfies analogous geometric bounds.

4.3 Case (C): Resonant distinct roots ($m := r_1 - r_2 \in \mathbb{N}$).

One always has a Frobenius series with exponent r_1 :

$$u_1(t) = t^{r_1} \sum_{n=0}^{\infty} c_n^{(1)} t^n.$$

Attempting a pure series with exponent r_2 fails at $n = m$ because $P(r_2 + m) = P(r_1) = 0$. Two subcases occur:

(C1) *No logarithm (exceptional compatibility)*. A certain linear compatibility condition at $n = m$ happens to hold, and one gets a pure series $u_2(t) = t^{r_2} \sum_{n=0}^{\infty} c_n^{(2)} t^n$.

Heuristically: remember from the discussion in class that the recurrence relation till $n = m - 1$ already defines the constants $c_0, c_1, \dots, c_{m-1}$. If $c_0 \neq 0$, then the recurrence relation obtained at the $n = m$ level is actually a constraint for the already defined $c_0, c_1, \dots, c_{m-1}$. (C1) describes the “somewhat unusual” situation where this constraint is satisfied.

(C2) *Logarithmic second solution (generic)*. Otherwise there is a second solution of the form

$$u_2(t) = u_1(t) \log t + t^{r_2} \sum_{n=0}^{\infty} d_n t^n,$$

with the coefficients d_n determined recursively. The $\log t$ term is forced exactly by the obstruction at the resonant index $n = m$. The idea is similar to the case described in Subsection 4.2.

Heuristically, this is the case where $c_0 = 0$, which in turn implies that $c_1 = c_2 = \dots = c_{m-1} = 0$. Then the recurrence condition at $n = m$ looks like $p(r_2 + m)c_m = 0$, that is, $0 \cdot c_m = 0$. In this case, one has the liberty to choose any value of c_m one wants, and continue the recursion.

4.4 Case (D): Complex conjugate roots $r, \bar{r}$ with $r = \alpha + i\beta$, $\beta \neq 0$.

(For real a_0, b_0 , nonreal roots come as a conjugate pair; note $r - \bar{r} = 2i\beta \notin \mathbb{Z}$, so this is nonresonant.) Let

$$U(t) = t^r \sum_{n=0}^{\infty} c_n t^n, \quad \overline{U(t)} = t^{\bar{r}} \sum_{n=0}^{\infty} \bar{c}_n t^n,$$

be the conjugate Frobenius pair (choose $c_0 = 1$ without loss of generality and observe the recursion preserves conjugation). Write $\phi(t) = \sum_{n \geq 0} c_n t^n = A(t) + iB(t)$ with A, B real-analytic and $A(0) = 1$,

²this is a special instance of what is known as the *Dominated Convergence theorem*. Feel free to look it up!

$B(0) = 0$. Then, for $t > 0$,

$$\begin{aligned} u_1(t) &= \operatorname{Re} U(t) = t^\alpha \left(A(t) \cos(\beta \log t) - B(t) \sin(\beta \log t) \right), \\ u_2(t) &= \operatorname{Im} U(t) = t^\alpha \left(A(t) \sin(\beta \log t) + B(t) \cos(\beta \log t) \right). \end{aligned}$$

These *real-valued* functions are linearly independent and form a fundamental system on $(0, R^*)$. No logarithmic terms occur in this case.

5 Wronskian and structural consequences.

Write (1) as $u'' + P(t)u' + Q(t)u = 0$ with $P(t) = a(t)/t$, $Q(t) = b(t)/t^2$. The Wronskian $W = u_1 u_2' - u_1' u_2$ satisfies

$$W'(t) = -P(t)W(t) \quad \Rightarrow \quad W(t) = C \exp\left(-\int^t \frac{a(s)}{s} ds\right).$$

Since $a(s) = a_0 + \sum_{k \geq 1} a_k s^k$,

$$\int^t \frac{a(s)}{s} ds = a_0 \log t + h(t), \quad h(t) \text{ analytic at } 0, \quad h(0) = 0.$$

Hence

$$W(t) = C t^{-a_0} e^{-h(t)} \sim C t^{-a_0} \quad (t \rightarrow 0^+).$$

6 Behaviour of solutions near $t = 0$

Let r_1, r_2 be the indicial roots. Then:

- If $\max\{\operatorname{Re} r_1, \operatorname{Re} r_2\} > 0$, there exists a nontrivial solution that *vanishes* at 0 (hence is bounded) of the form $t^{r_j}(1 + o(1))$ for a root with positive real part.
- If $\min\{\operatorname{Re} r_1, \operatorname{Re} r_2\} < 0$, at least one nontrivial solution *blows up* at 0.
- If both $\operatorname{Re} r_j < 0$, *every* nontrivial solution blows up.
- If some $\operatorname{Re} r_j = 0$, boundedness depends on resonance and coefficients (a bounded nontrivial solution may or may not exist).

7 Real solutions on $t < 0$.

It is often preferable to treat the intervals $(0, R^*)$ and $(-R^*, 0)$ separately.

Set $s = -t > 0$ and define $v(s) = u(-s)$. Then $u'(t) = -v'(s)$, $u''(t) = v''(s)$, and (1) becomes

$$s^2 v''(s) + s a(-s) v'(s) + b(-s) v(s) = 0.$$

Since $a(-s) = \sum_{k \geq 0} a_k (-1)^k s^k$ and $b(-s) = \sum_{k \geq 0} b_k (-1)^k s^k$ are analytic on $s \in (0, R^*)$, the same Frobenius procedure applies with the *same* indicial polynomial $P(x)$ (only a_0, b_0 matter). Thus there exist solutions on $(-R^*, 0)$ of the form

$$u_j^-(t) = |t|^{r_j} \sum_{n=0}^{\infty} \tilde{c}_n^{(j)} |t|^n \quad \text{and, in the resonant/double cases,} \quad u_2^-(t) = u_1^-(t) \log |t| + |t|^{r_2} \sum_{n=0}^{\infty} \tilde{d}_n |t|^n,$$

with real coefficients (when a, b are real). Here one simply solves the recursions with the series of $a(-s), b(-s)$ (equivalently, replace a_j and b_j by $(-1)^j a_j$ and $(-1)^j b_j$ in the sums for the recursion), and then writes the answers in the variable $|t|$.